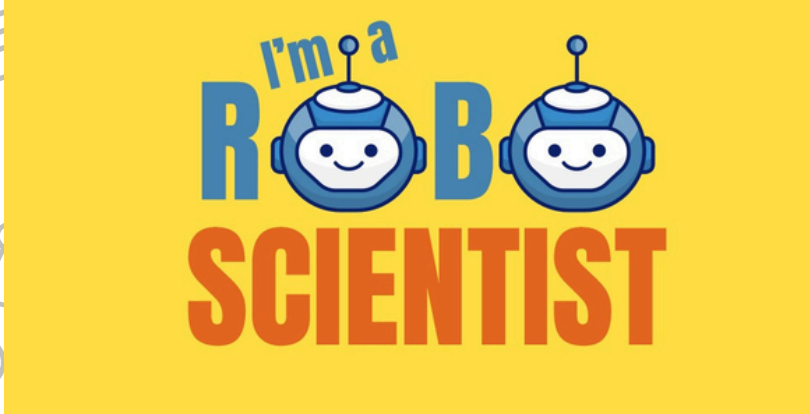
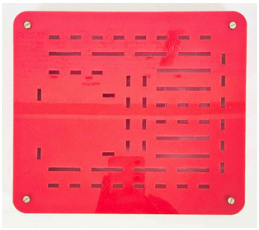
ScienceUtsav®
Science is awesome



Basics Of Light and Sound Circuits

Session 1

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2


3


4


5

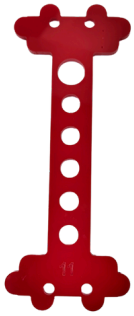

6


7


8


9


10


11


1

Spacers

2

L Clamp

3

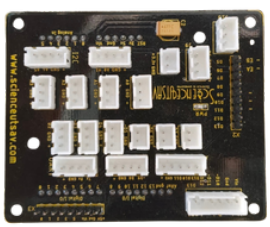
Spacer Screws

4

Nuts

5

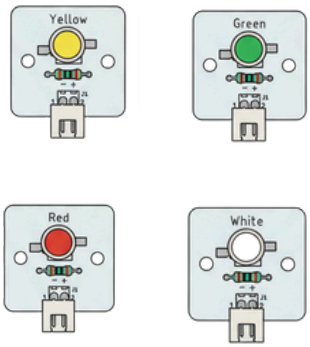
Screw

6

Arduino With Shield

7

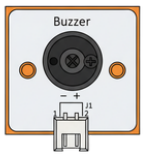
Arduino cable

ELECTRONIC COMPONENTS



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



BUZZER PCB

A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



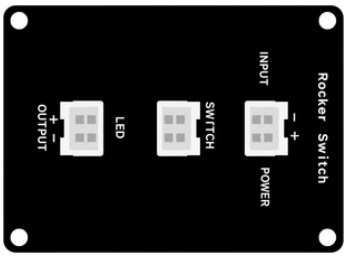
ROCKER SWITCH

A rocker switch is an electrical device that allows you to turn a circuit ON or OFF by pressing one side of the switch. It is frequently utilized in electronic projects and appliances for convenient power control.



Push Switch

A push switch is a button-style switch that operates only when pressed.
It provides temporary input in circuits, such as starting a device or triggering an action.



rocker switch PCB

A rocker switch PCB is a small circuit board designed to hold and connect a rocker switch easily in electronic projects.
It provides proper connections and makes the switch installation neat, stable, and easy to use.

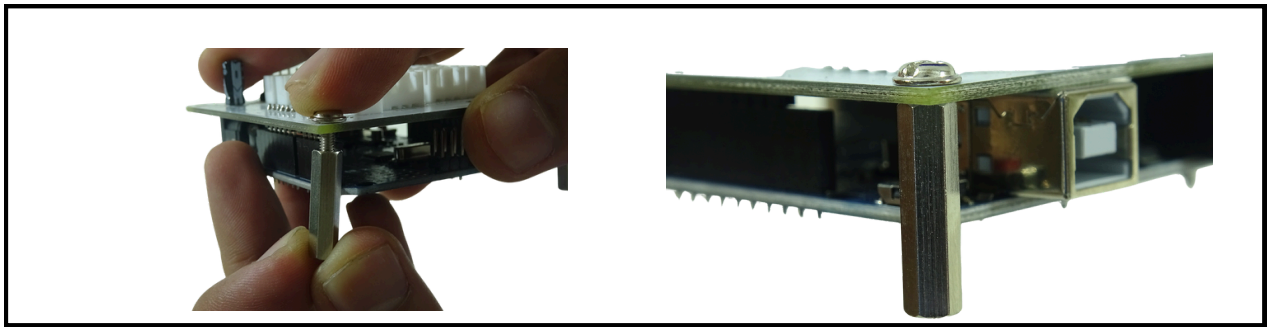


2 PIN RMC

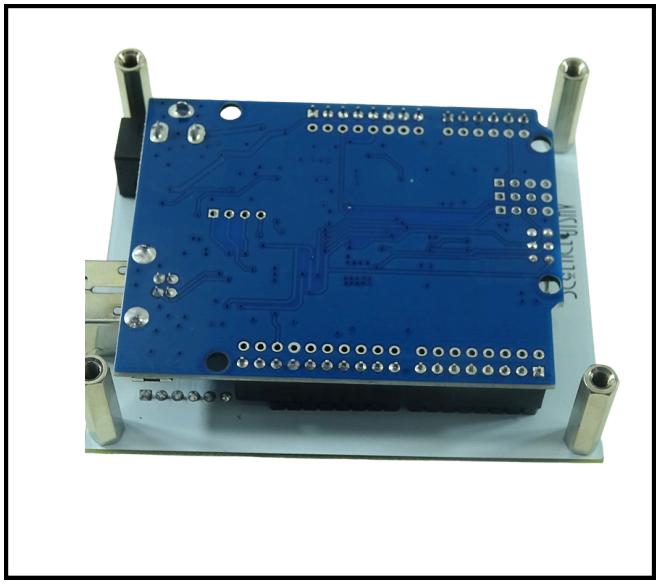
Used to connect the 1-watt LEDs and the buzzer.

Procedure :

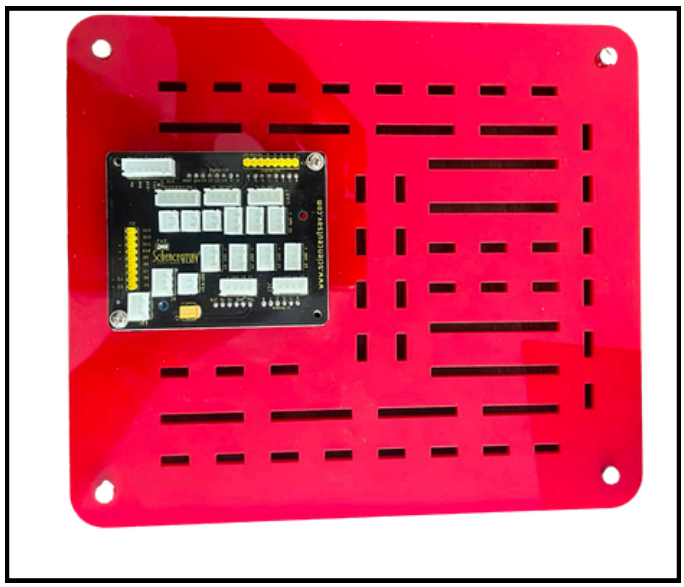
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

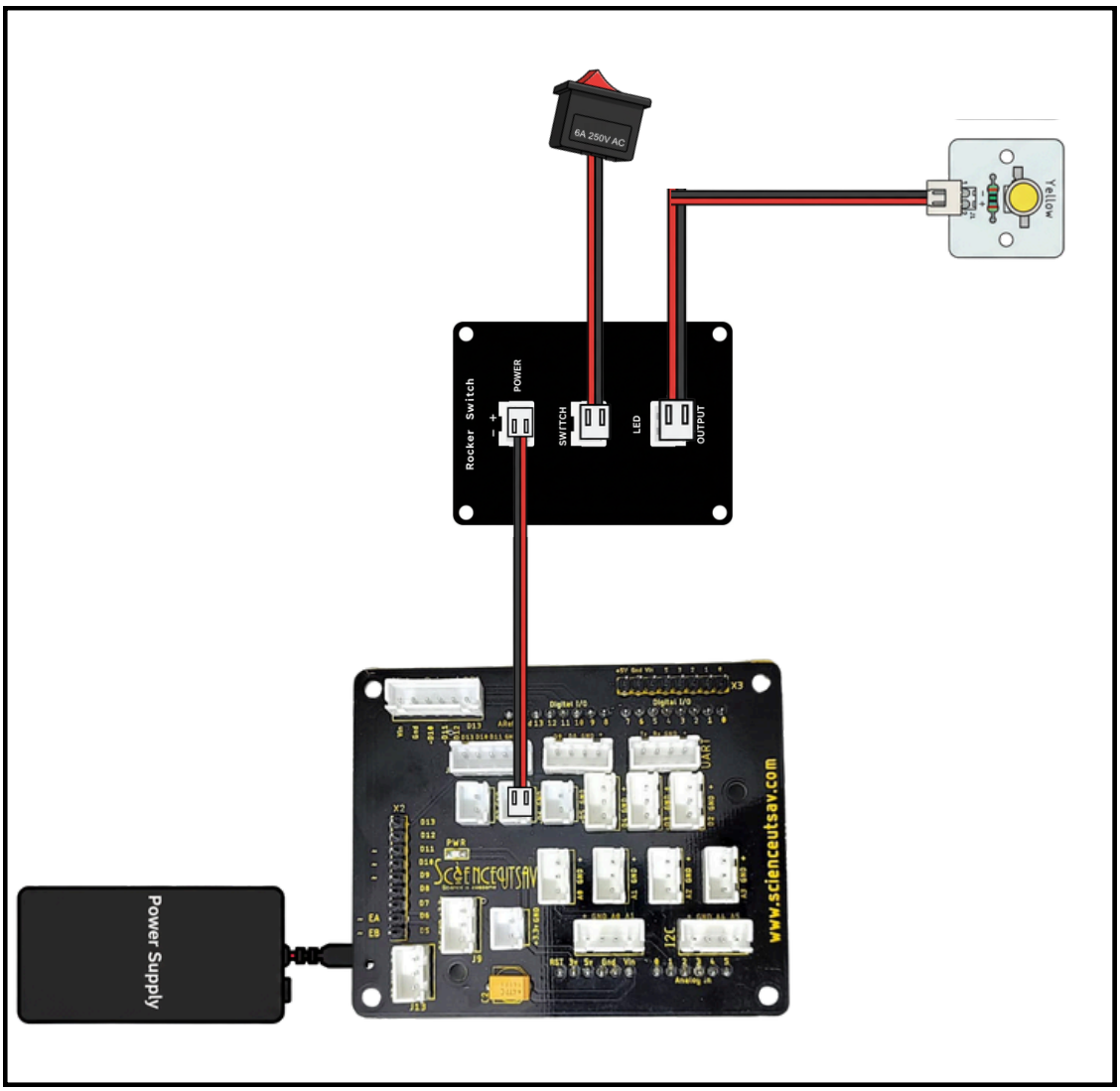


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

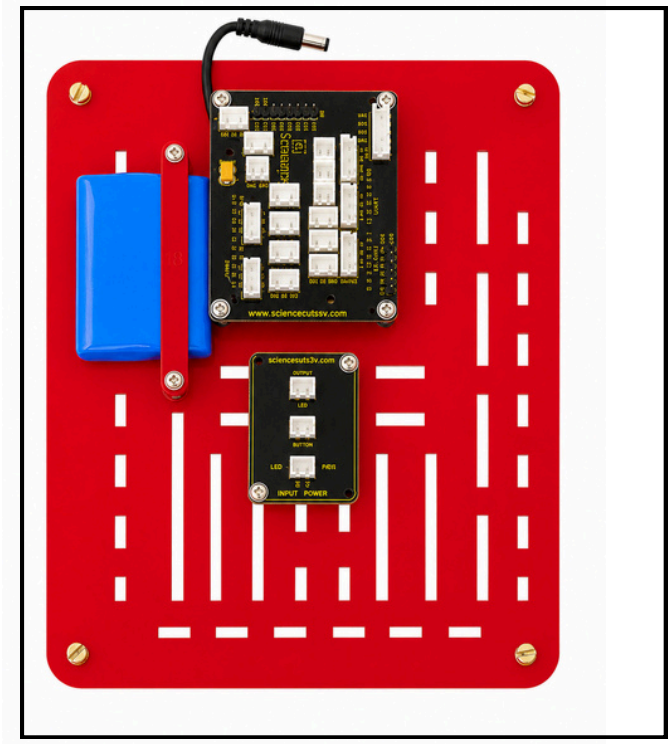
The circuit diagram of the project is given below



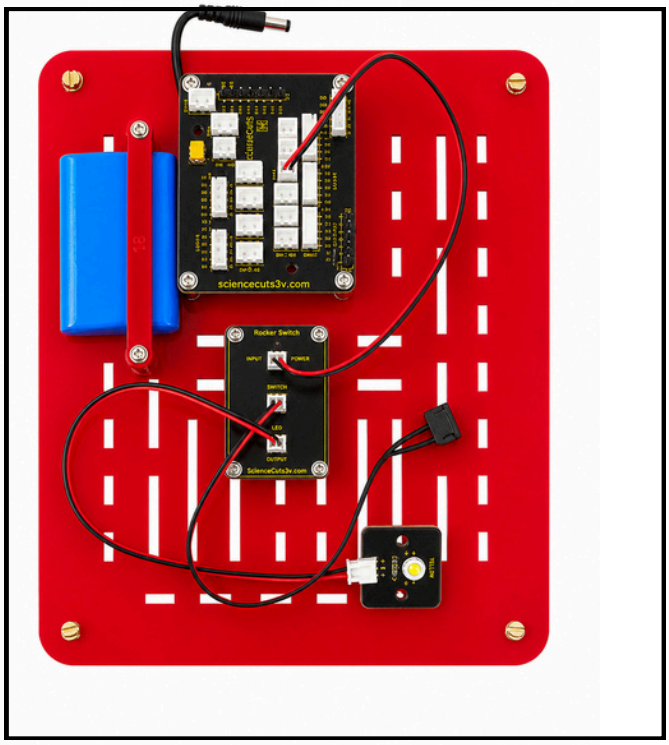
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- LED, rocker switch

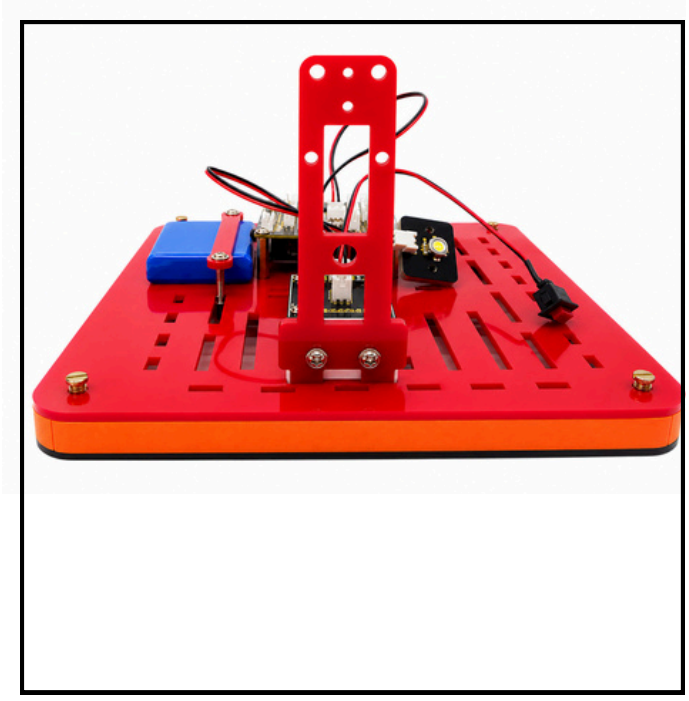
Step1- Fix the rocker switch PCB to the board



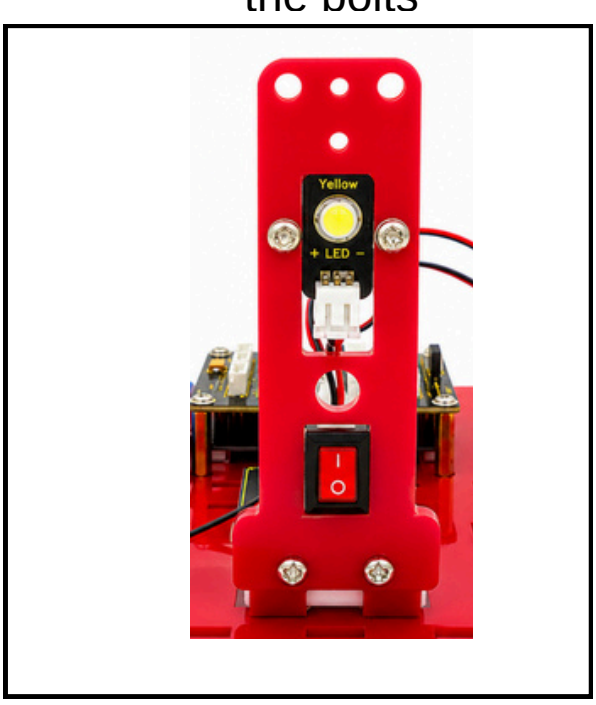
Step 2- Connect the rocker switch and the LED to the PCB



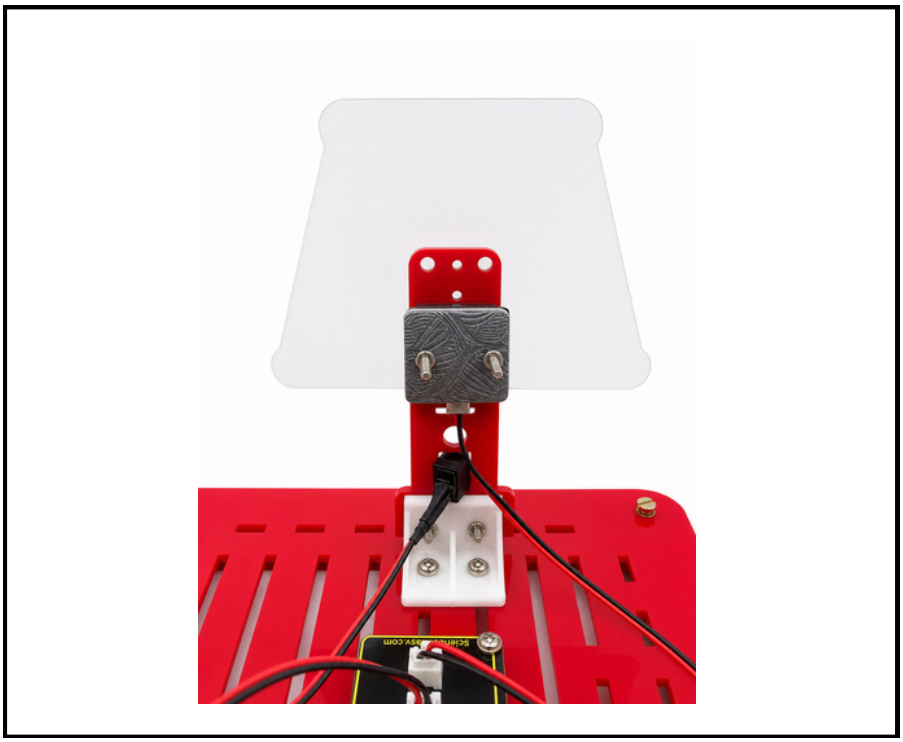
Step 3- Attach the switch holder as shown

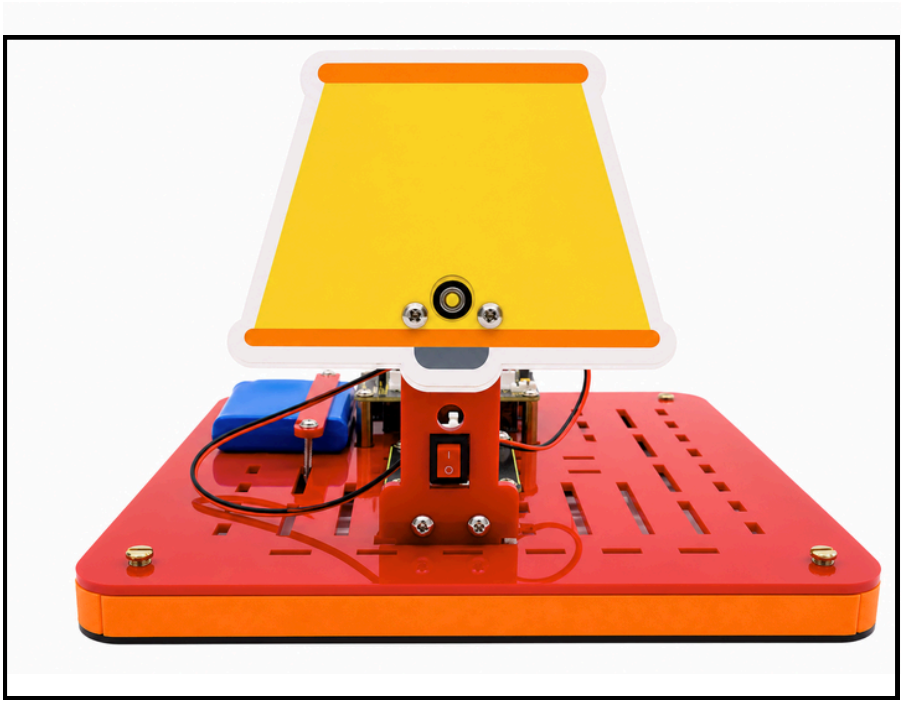


Step 4- Attach the rocker switch and the led to the holder and secure them with the bolts



Step 5- Place the template on the stand and secure it with the bolts





Your project is now ready for exploration!

code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

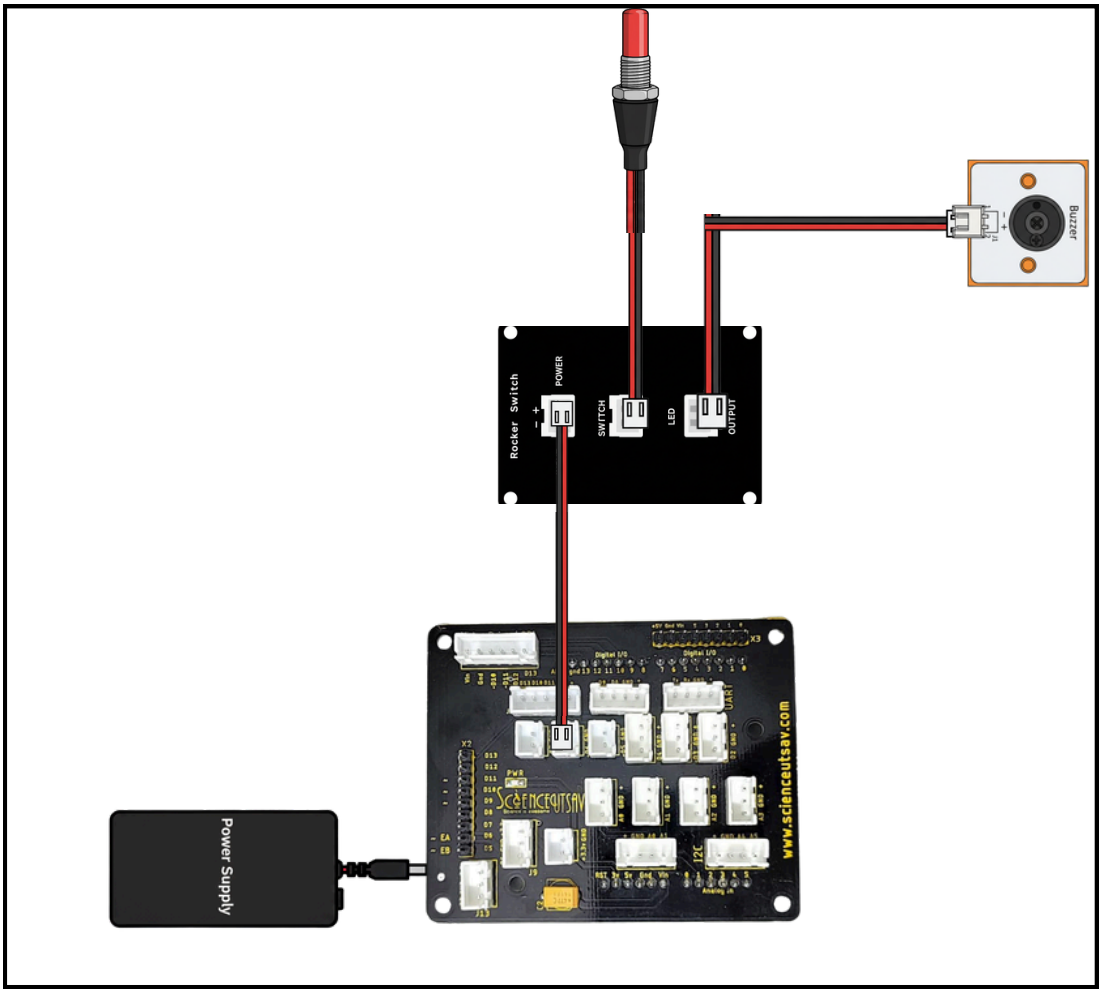
const int led = 6;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
}
```

BEGINNER PROJECT

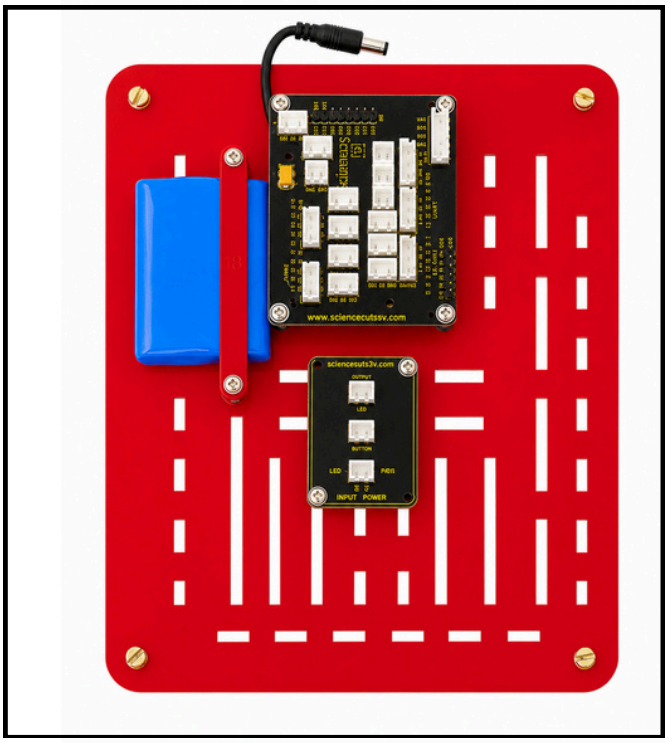
The circuit diagram of the project is given below



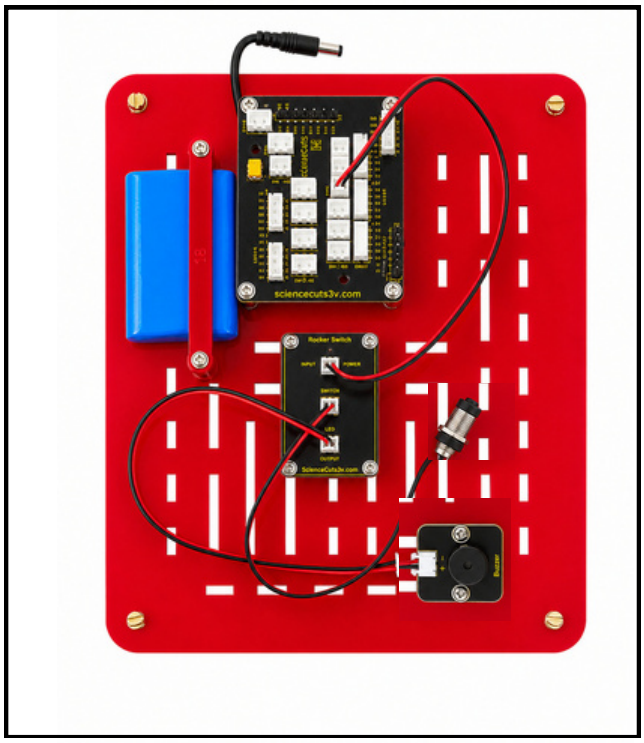
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- Buzzer, push button

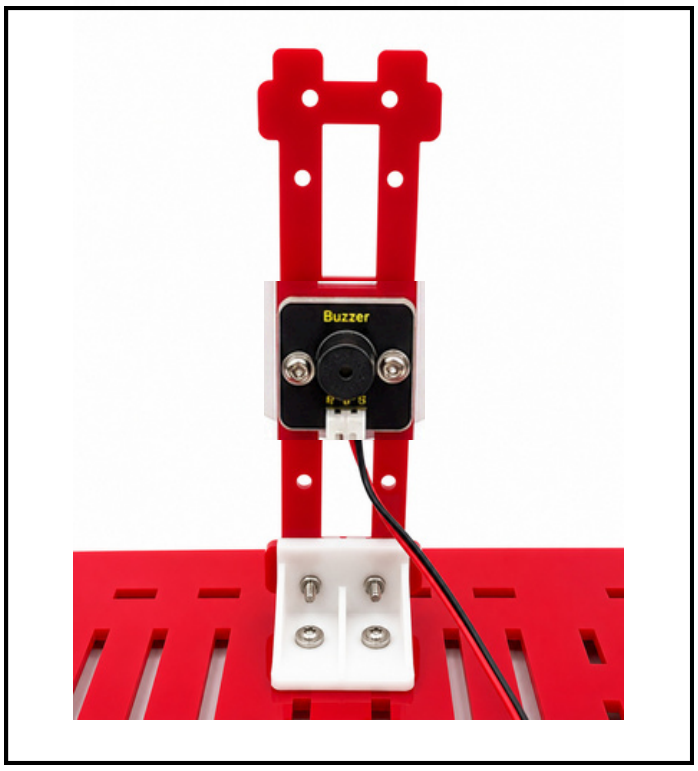
Step1- Fix the rocker switch PCB to the board



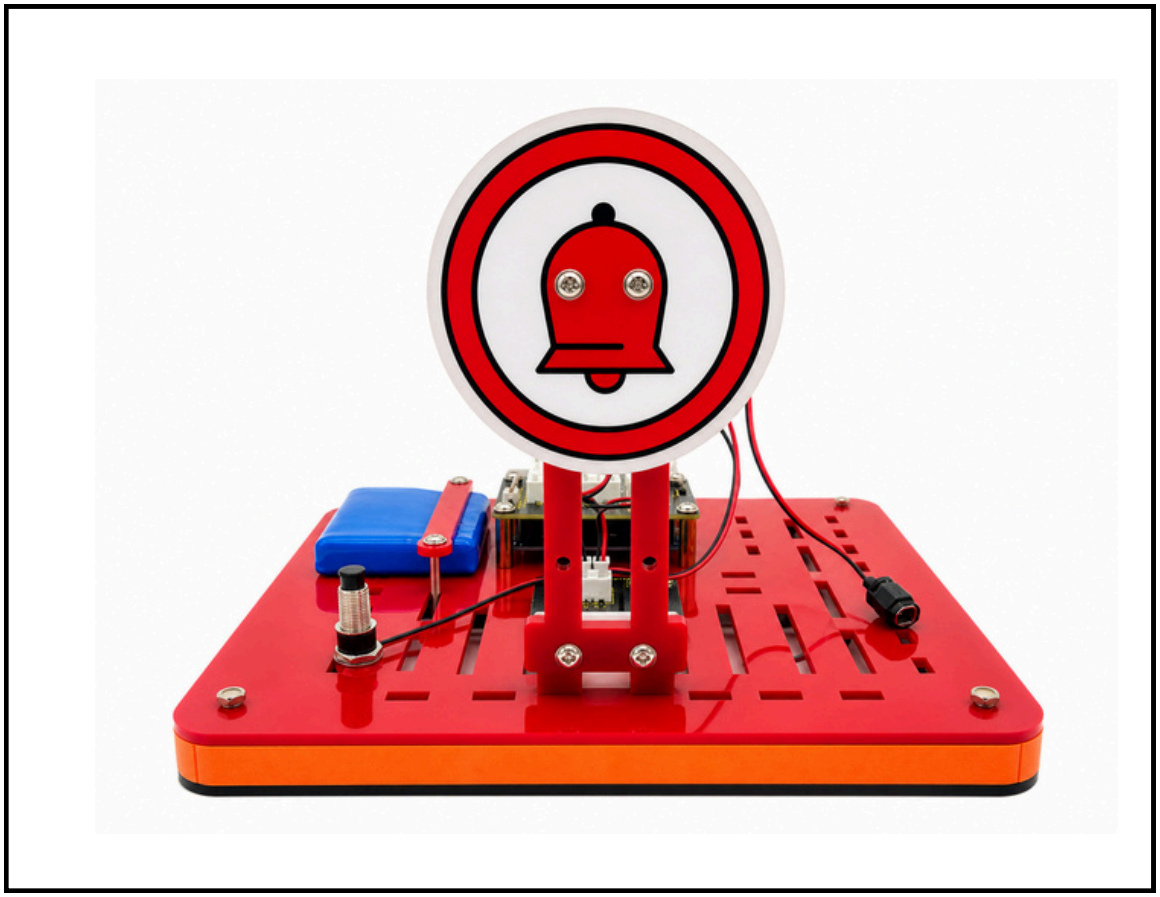
Step 2- Connect the push button and the buzzer to the PCB



Step 3- Attach the stand and connect the buzzzer to it. Secure it with bolt



Step 4- Attach the template using the bolts as shown



Your alarm is now ready to ring!

code

```
/*
Simple Alarm Sound
Buzzer connected to pin 6
*/

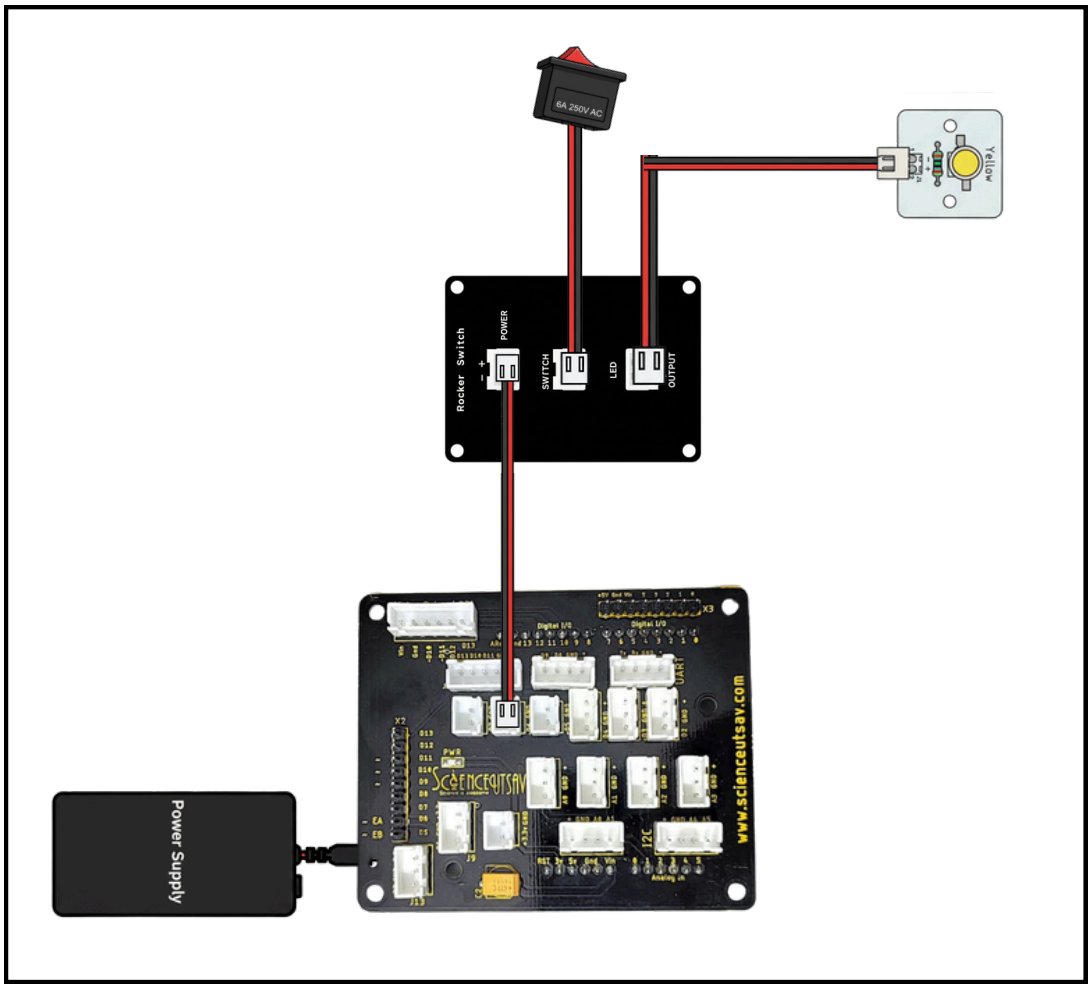
const int buzzer = 6;

void setup() {
  pinMode(buzzer, OUTPUT);
}

void loop() {
  digitalWrite(buzzer,HIGH);
}
```

INTERMEDIATE PROJECT

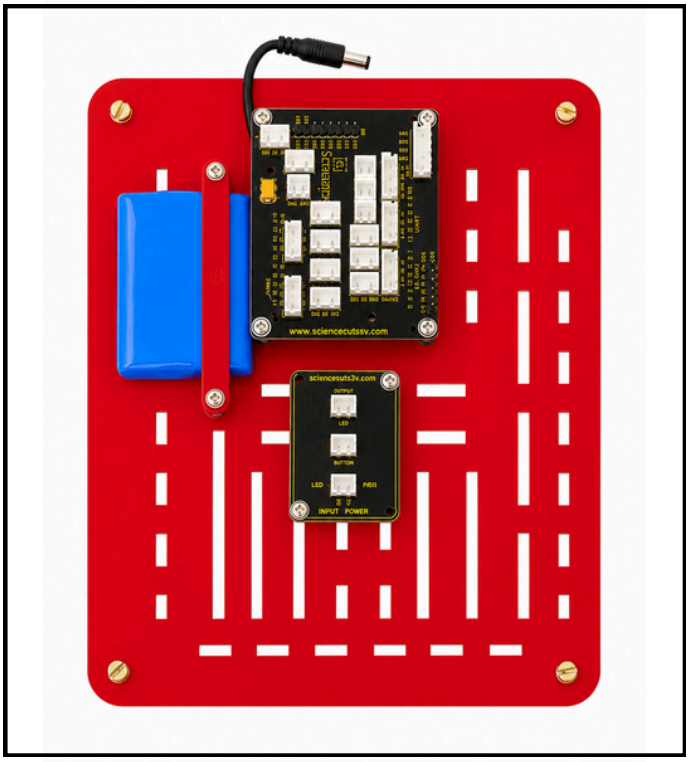
The circuit diagram of the project is given below



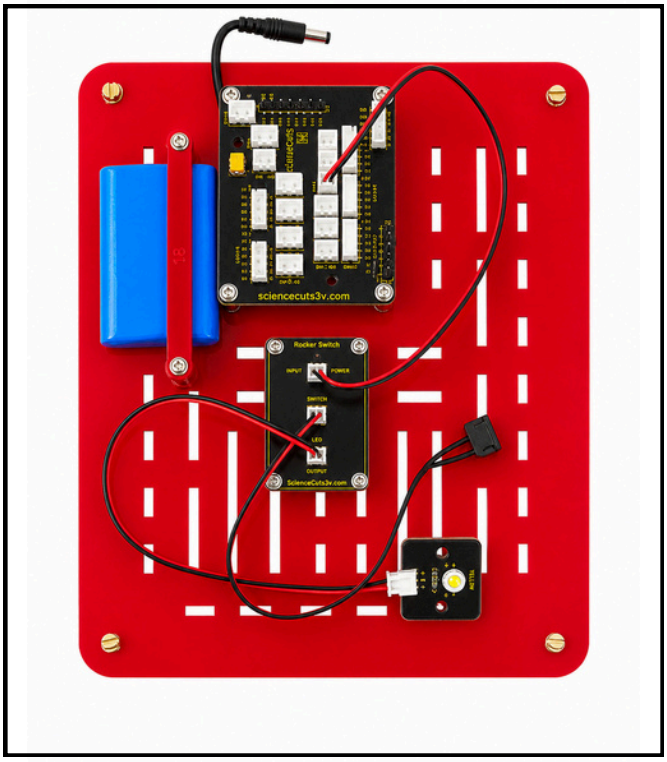
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- LED, rocker switch

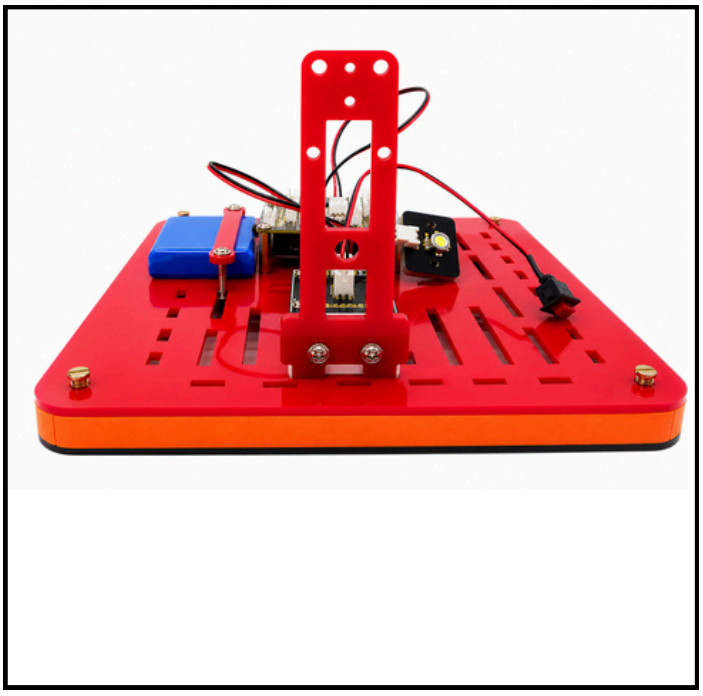
Step1- Fix the rocker switch PCB to the board



Step 2- Connect the rocker switch and the LED to the PCB



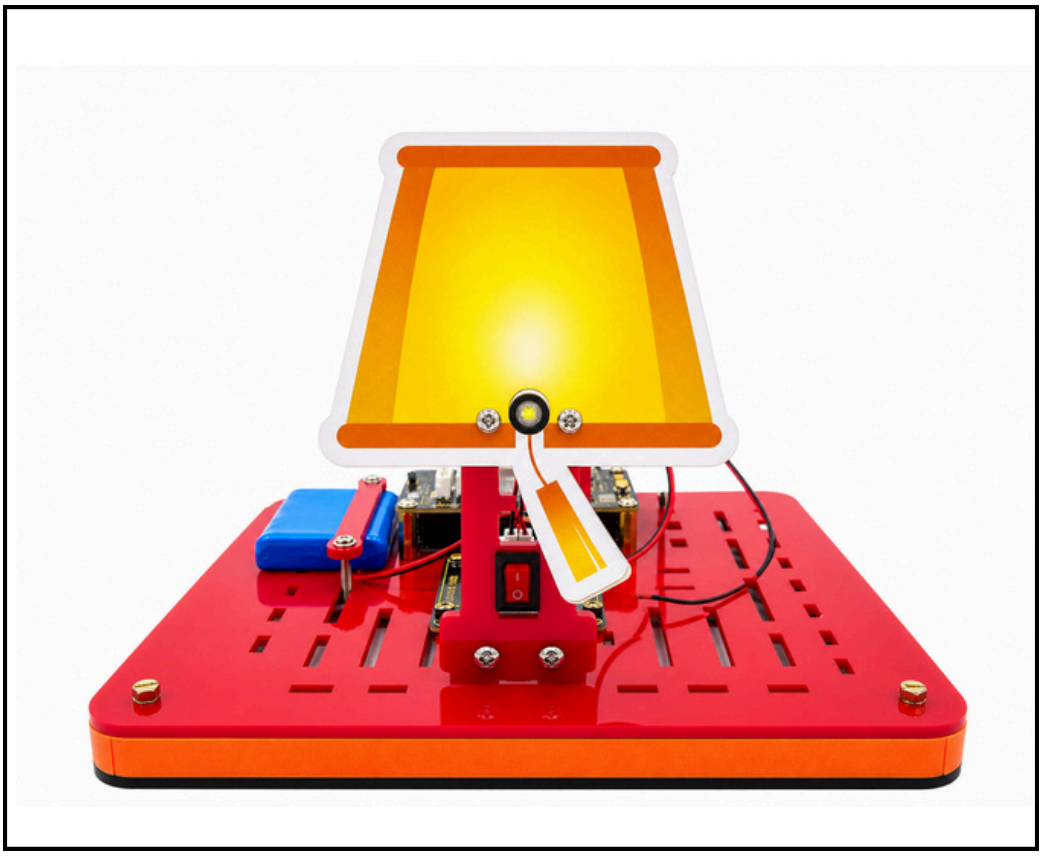
Step 3- Attach the switch holder as shown



Step 4- Attach the rocker switch and the led to the holder and secure them with the bolts



Step 6- Attach the template to the stand and secure it with bolts



Your lantern is now ready to soar high into the sky!

code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

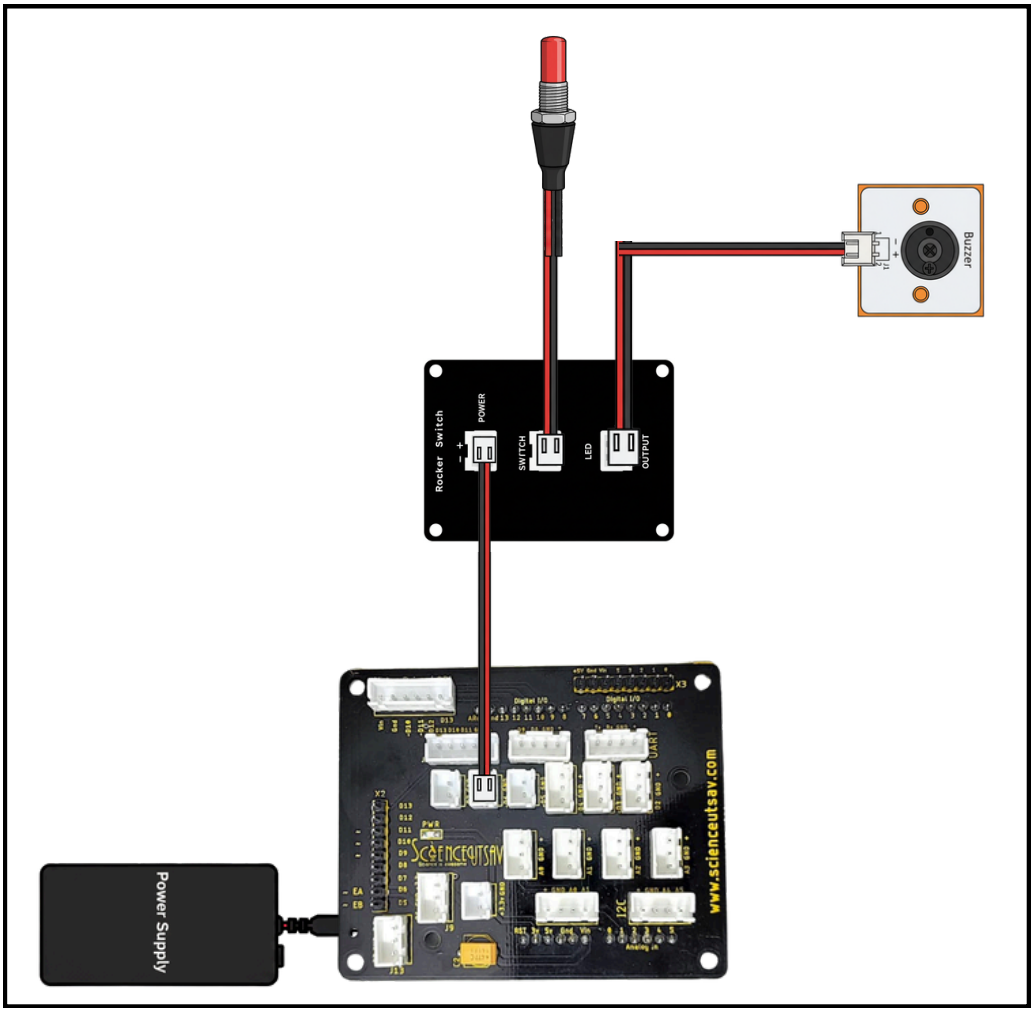
const int led = 6;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
}
```

INTERMEDIATE PROJECT

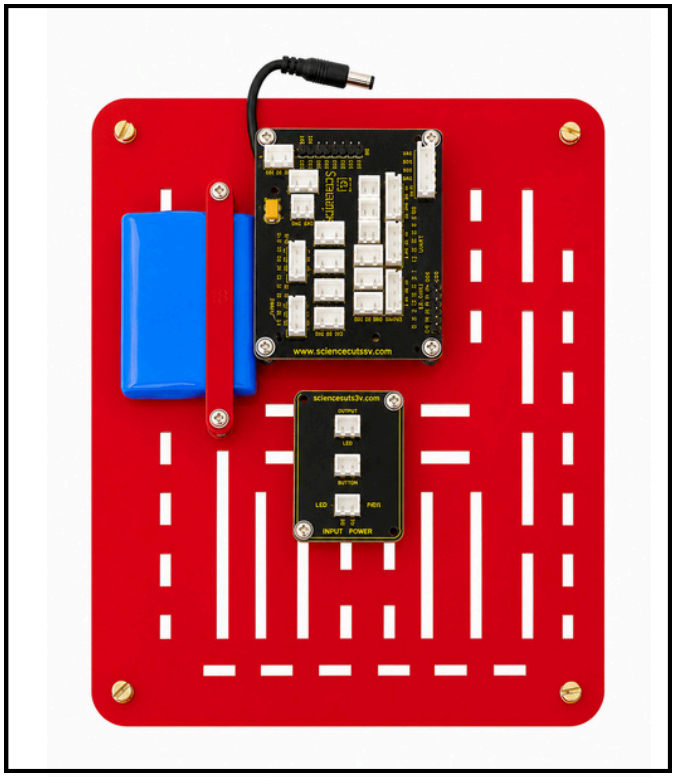
The circuit diagram of the project is given below



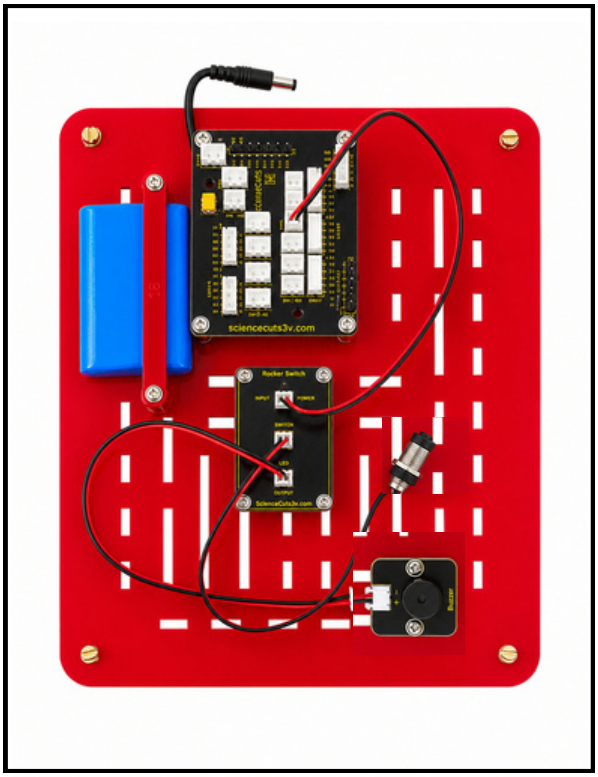
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- Buzzer, push button

Step 1- Fix the rocker switch PCB to the board



Step 2- Connect the push button and the buzzer to the PCB



Step 3- Attach the template to the clamp stand and secure it with bolts.



Step 4- Attach it to the board as shown.



code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

const int buzzer = 6;

void setup() {

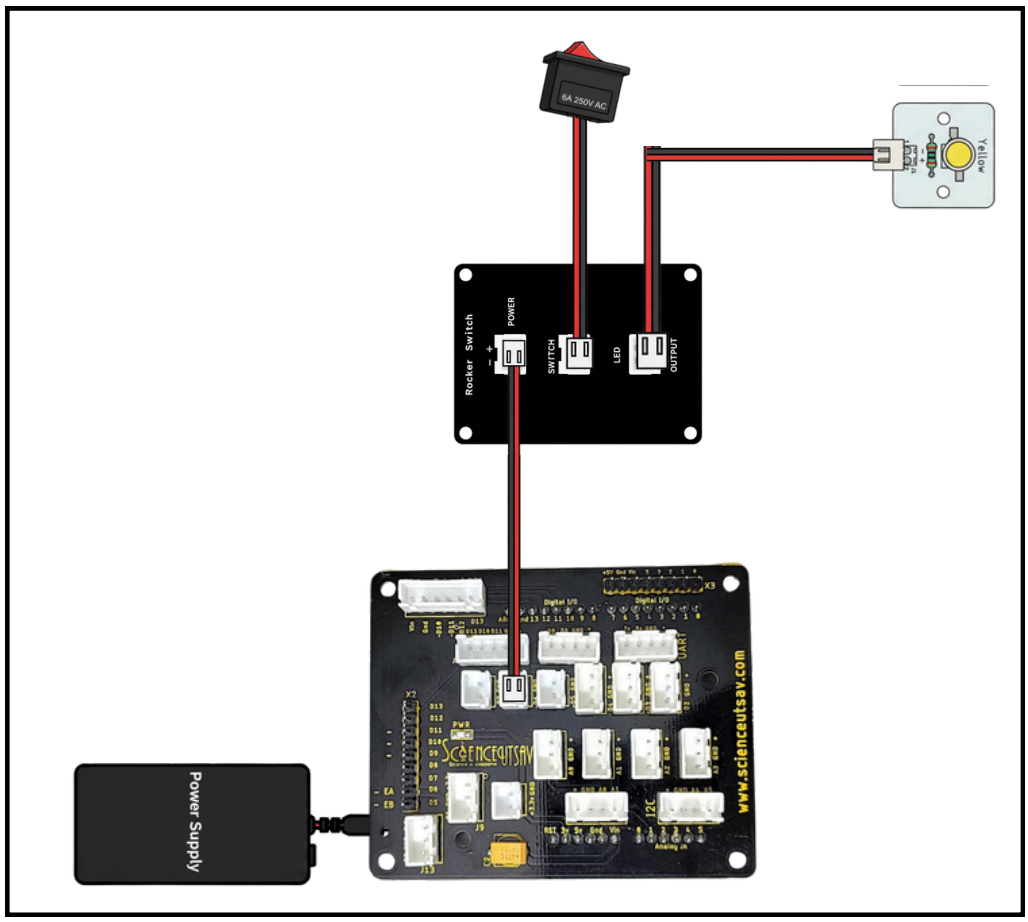
  pinMode(buzzer, OUTPUT);
}

void loop() {
  digitalWrite(buzzer, HIGH);

}
```

ADVANCE PROJECT

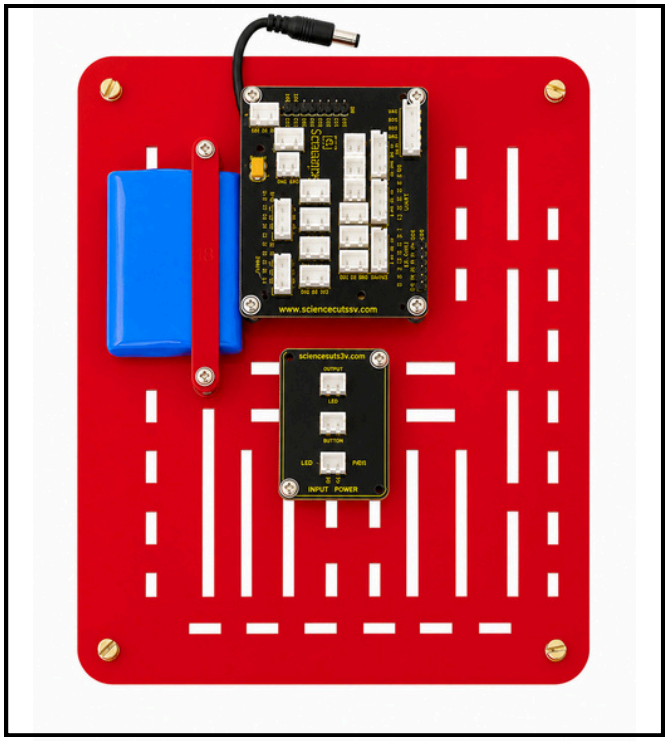
The circuit diagram of the project is given below



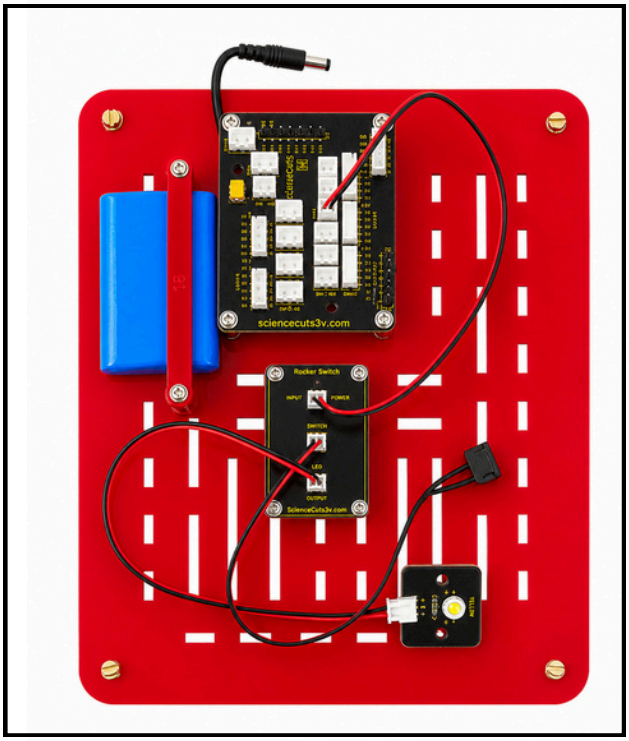
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- LED, rocker switch

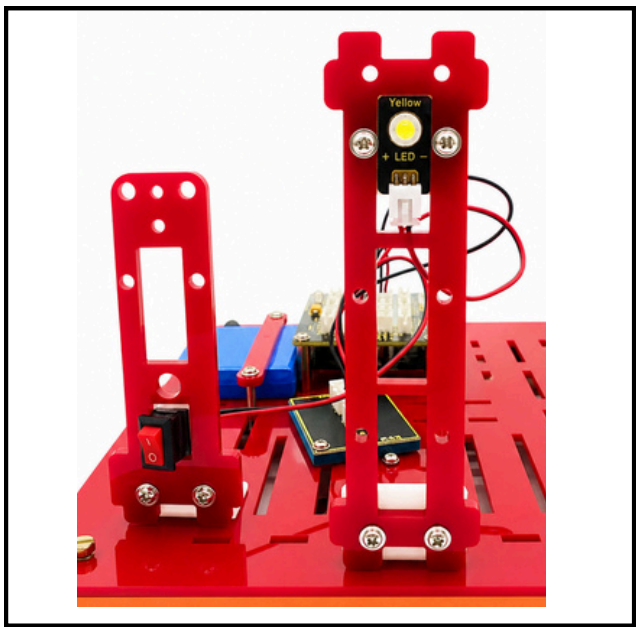
Step1- Fix the rocker switch PCB to the board



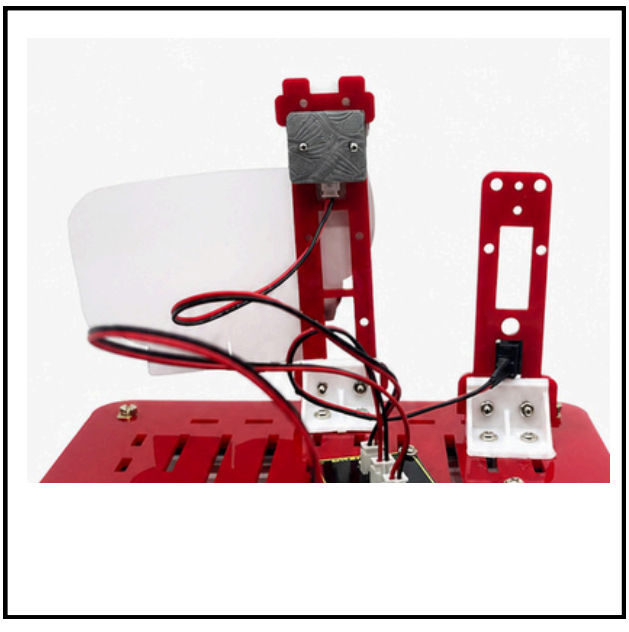
Step 2- Connect the rocker switch and the LED to the PCB



Step 3- Attach both the stands and secure the LED and rocker switch with the bolts as shown



Step 4- Attach the template as shown





Your messenger is now ready to light up the conversation!

code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

const int led = 6;

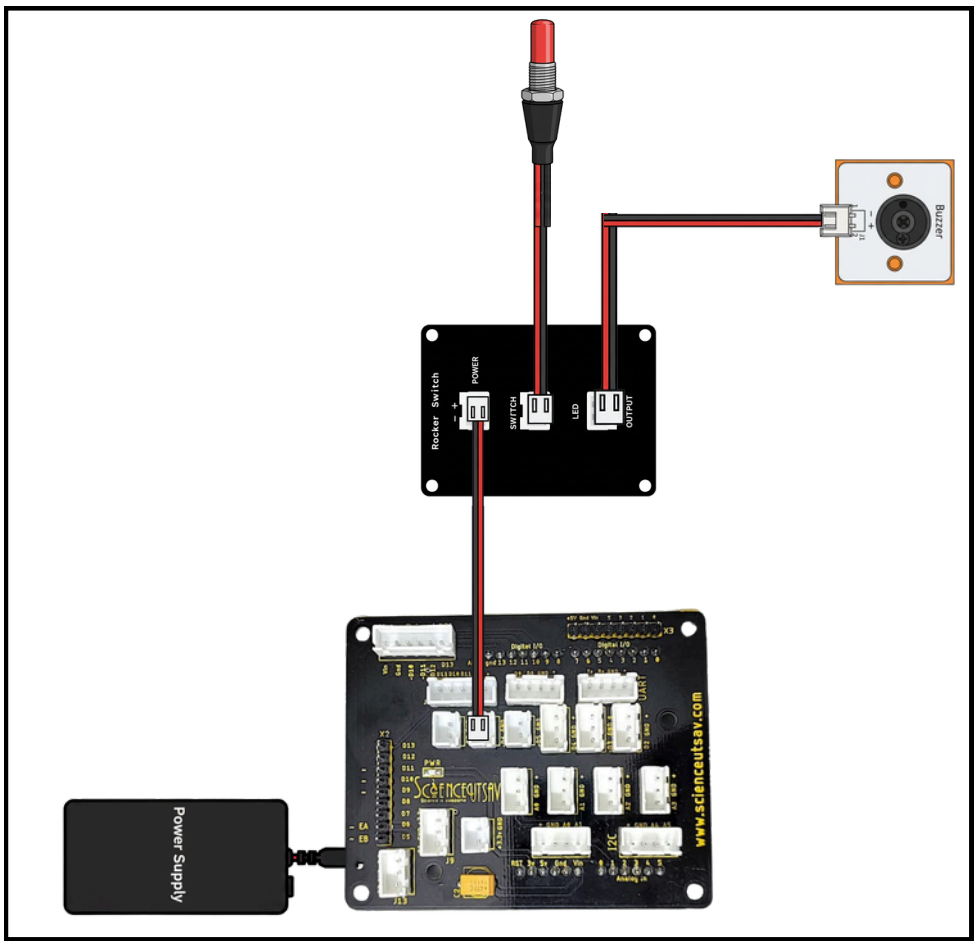
void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
  delay(1000);
  digitalWrite(led, LOW);
  delay(1000);

}
```

ADVANCE PROJECT

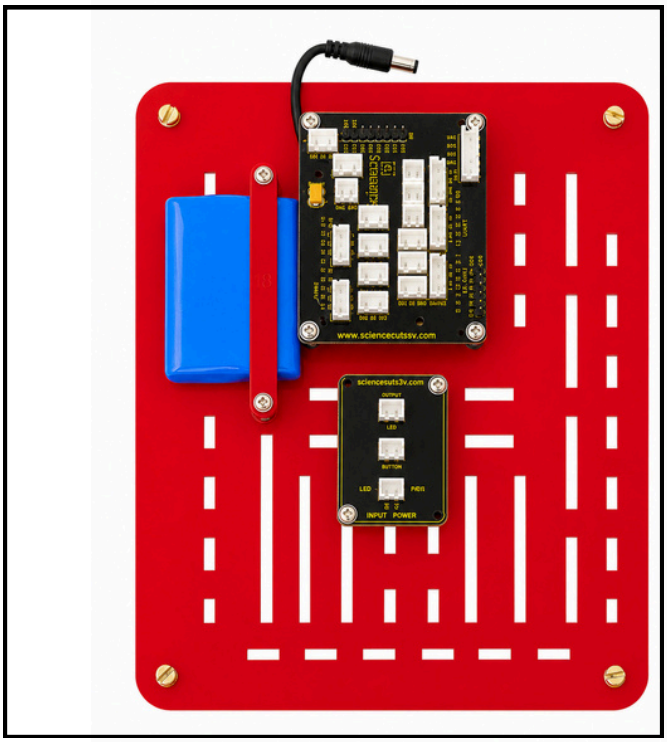
The circuit diagram of the project is given below



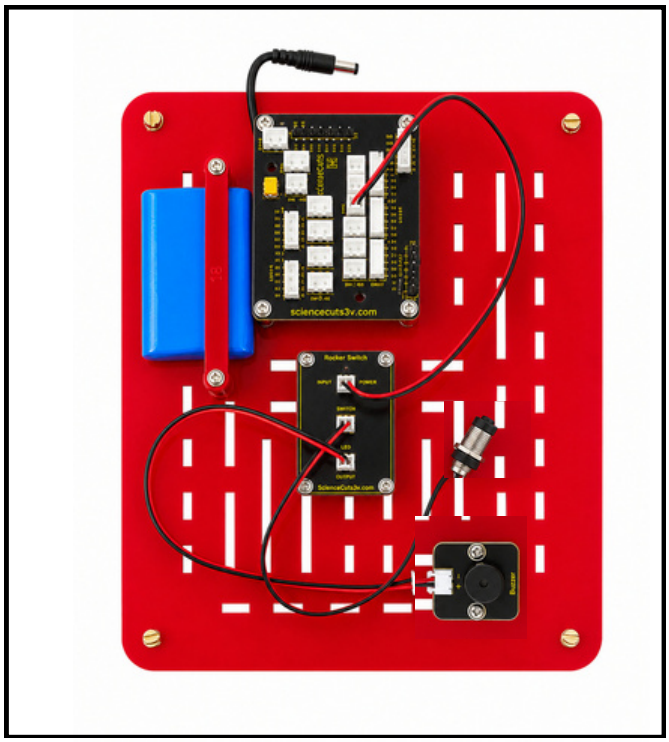
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- Buzzer, push button

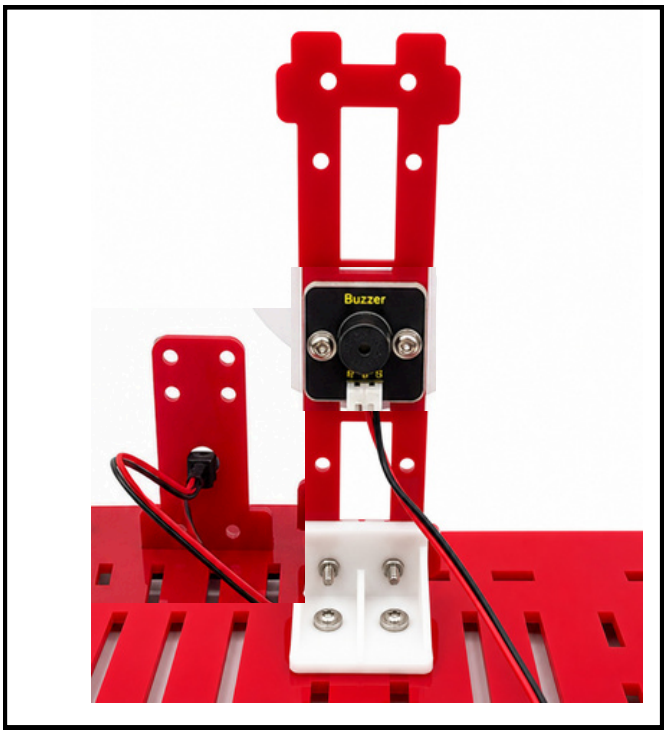
Step1- Fix the rocker switch PCB to the board



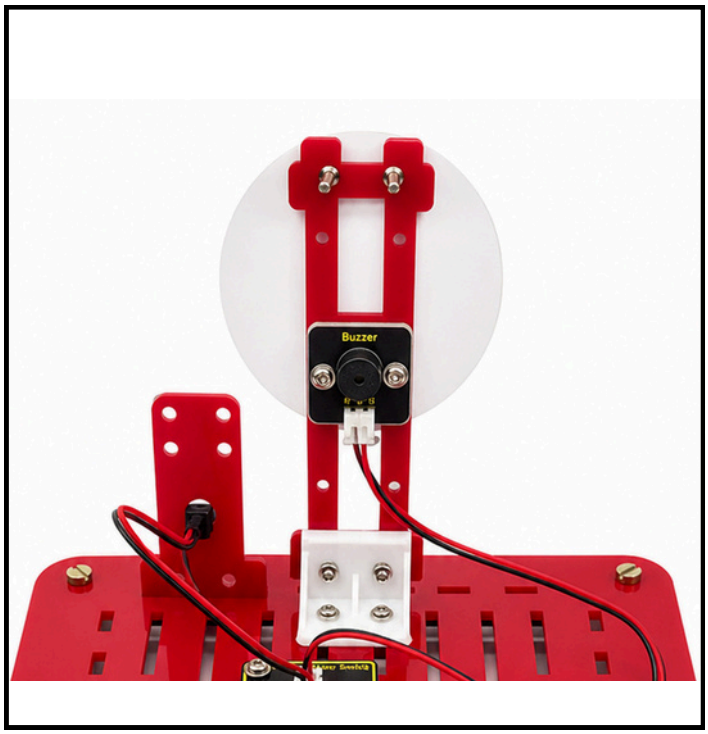
Step 2- Connect the push button and the buzzer to the PCB

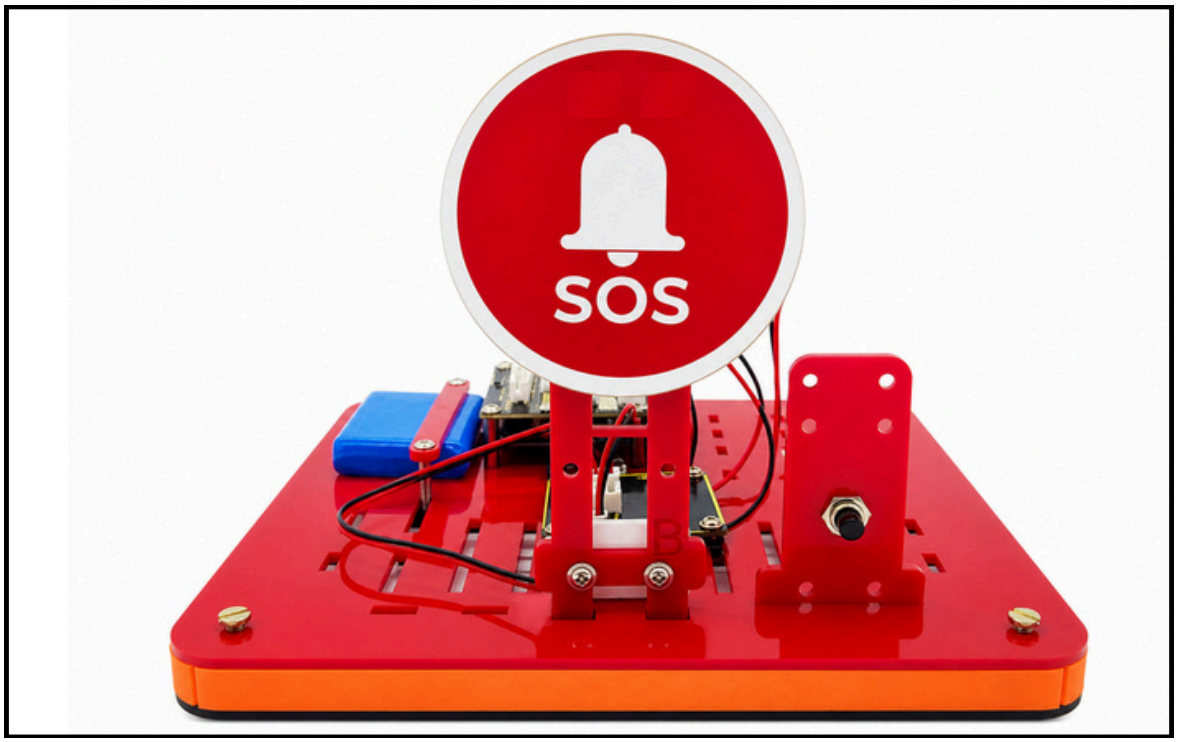


Step 3- Attach the stands. Connect the buzzer and push switch to it. Secure it with bolt



Step 4- Attach the template and secure it with bolts as shown





Your SOS is now ready to trigger emergencies!

code

```
/*
SOS Alarm Sound
Pattern: ... --- ...
Buzzer connected to pin 6
*/

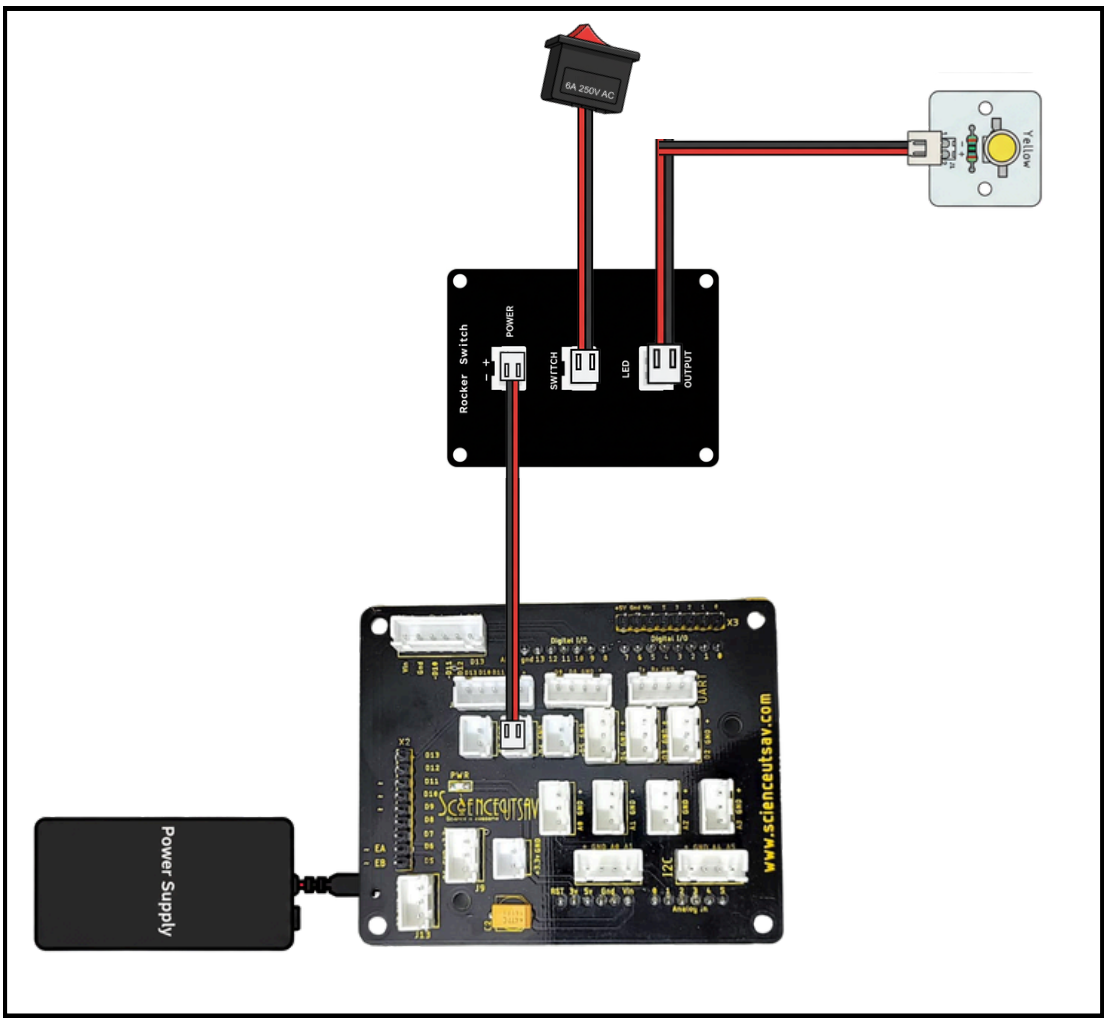
const int buzzer = 6;

void setup() {
  pinMode(buzzer, OUTPUT);
}

void loop() {
  digitalWrite(buzzer,HIGH);
}
```

EXPERT PROJECT

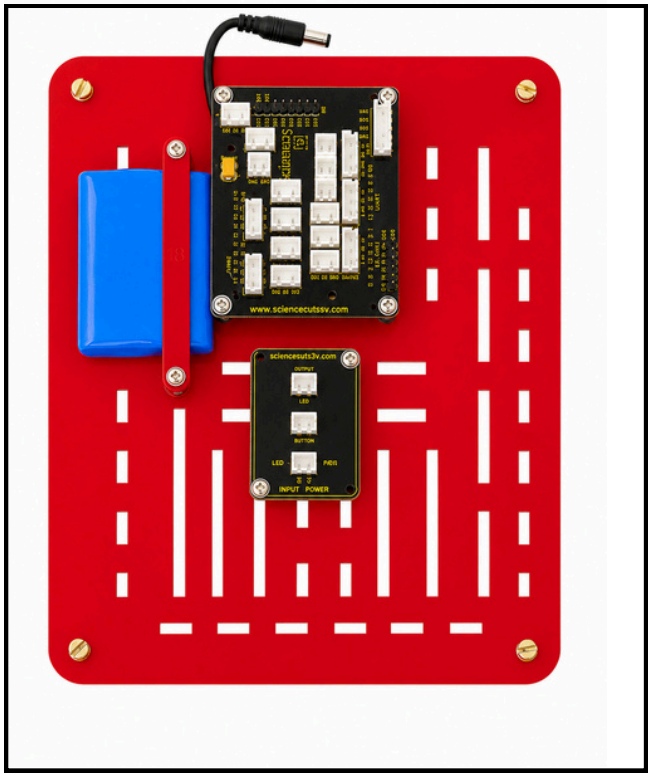
The circuit diagram of the project is given below



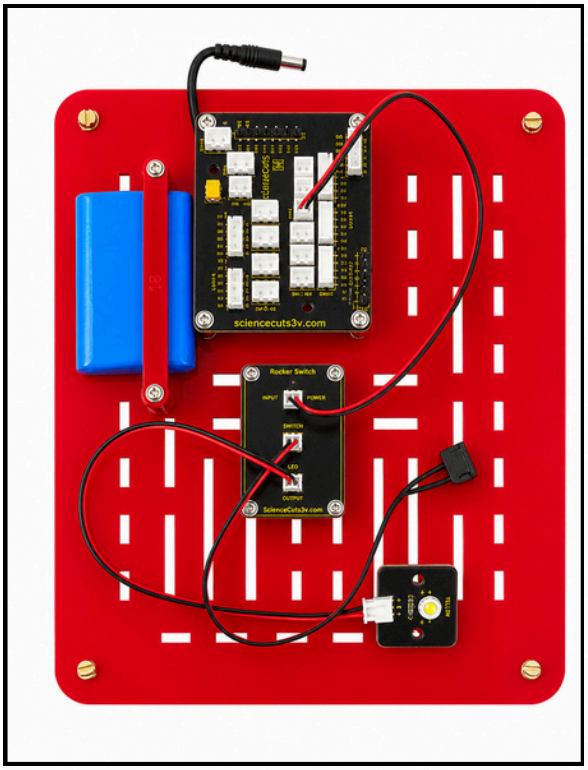
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- LED, rocker switch

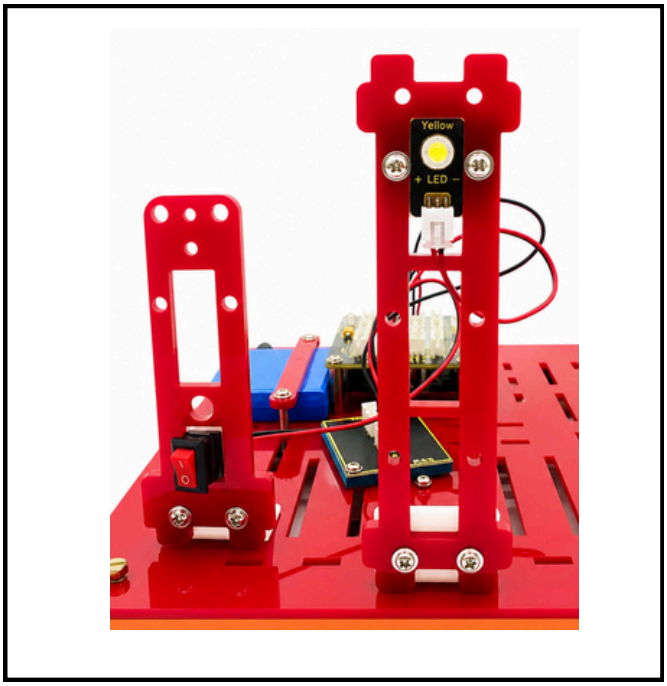
Step1- Fix the rocker switch PCB to the board



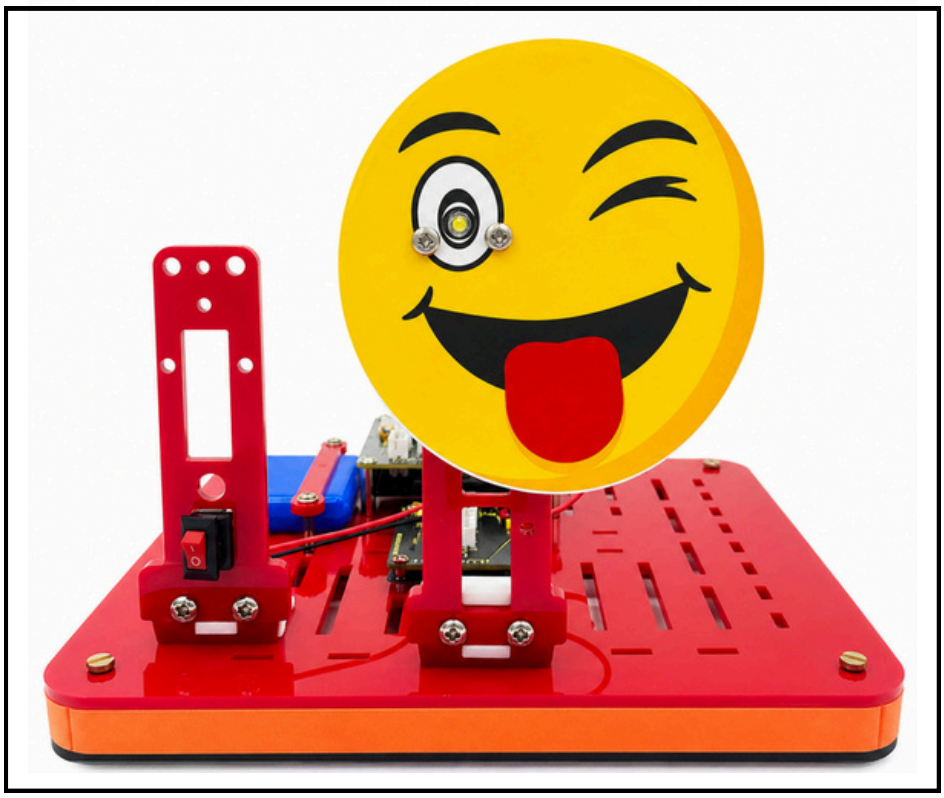
Step 2- Connect the rocker switch and the LED to the PCB



Step 3- Attach both the stands and secure the LED and rocker switch with the bolts as shown



Step 4- Attach the template as shown and secure it with bolts.



Your smiley is now ready to shine!

code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

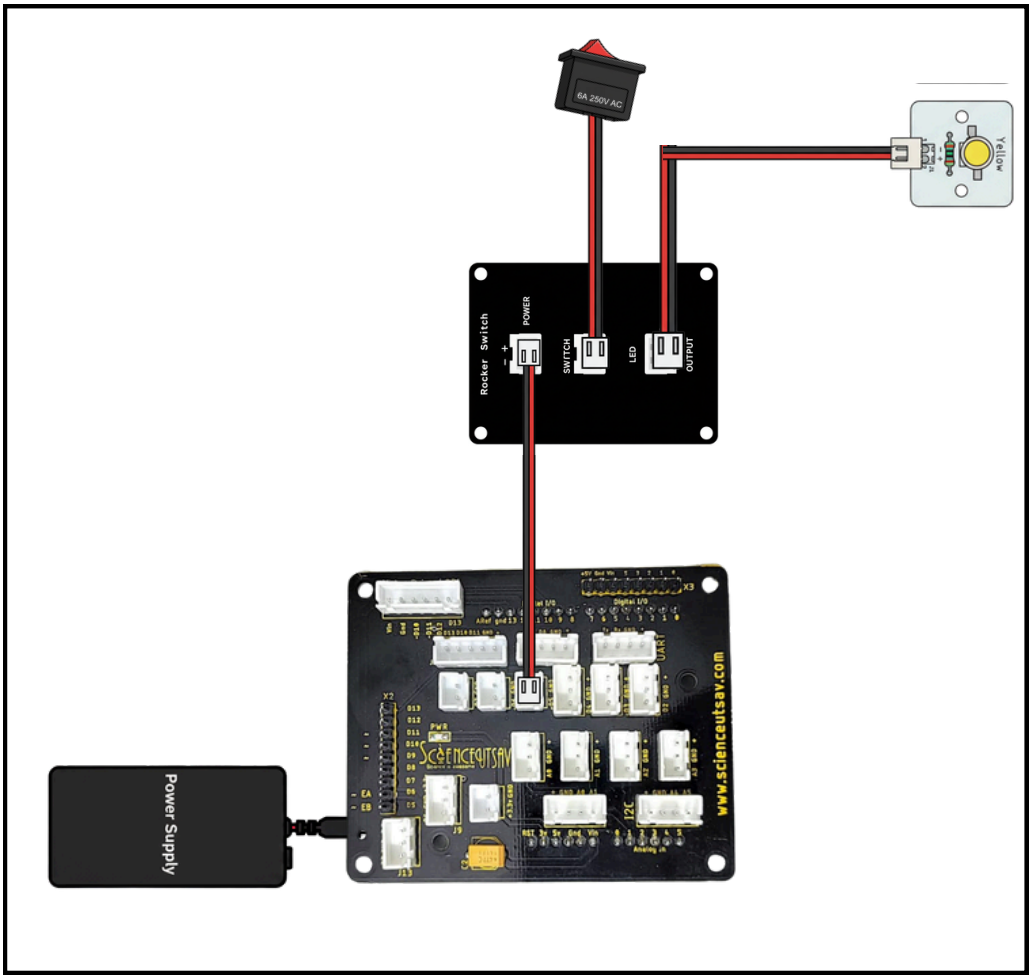
const int led = 6;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
  delay(500);
  digitalWrite(led, LOW);
  delay(500);
}
```

EXPERT PROJECT

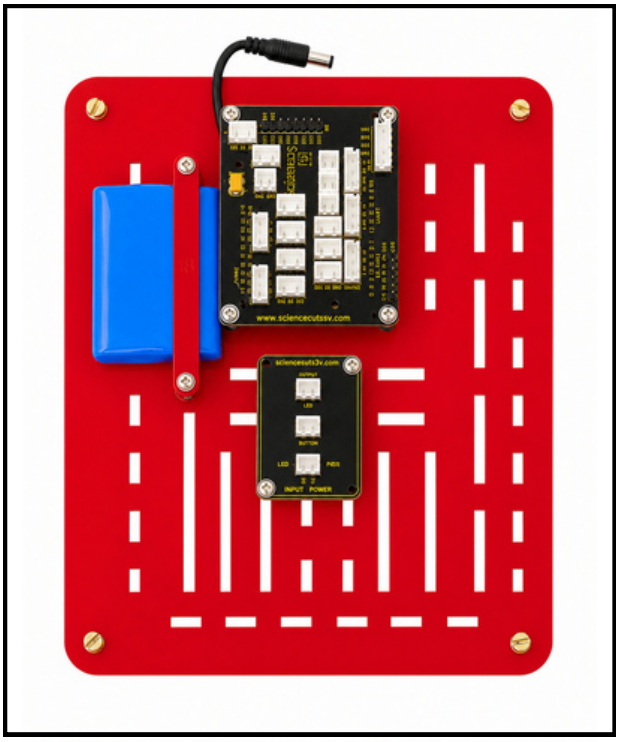
The circuit diagram of the project is given below



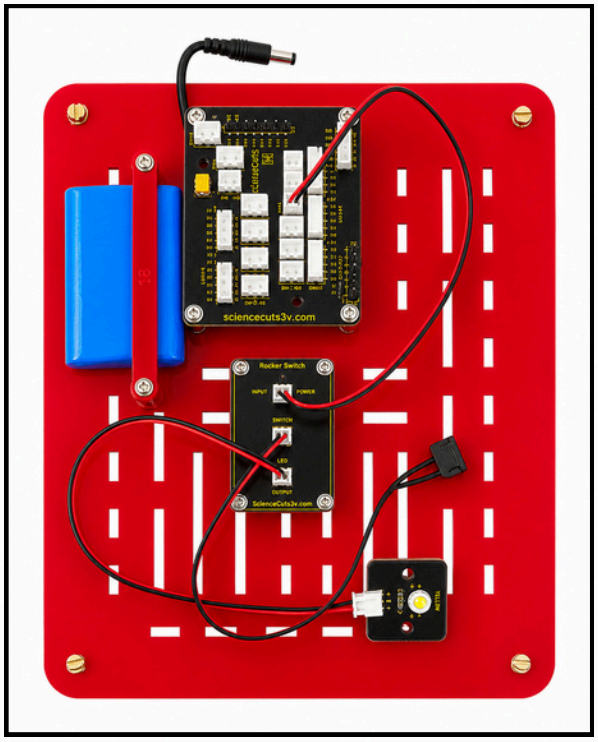
CONNECTION :

1. D6- Rocker switch PCB
2. Rocker switch PCB- LED, rocker switch

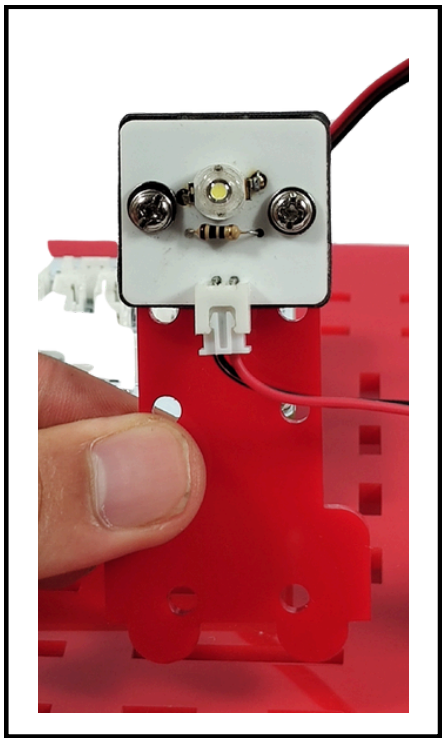
Step1- Fix the rocker switch PCB to the board

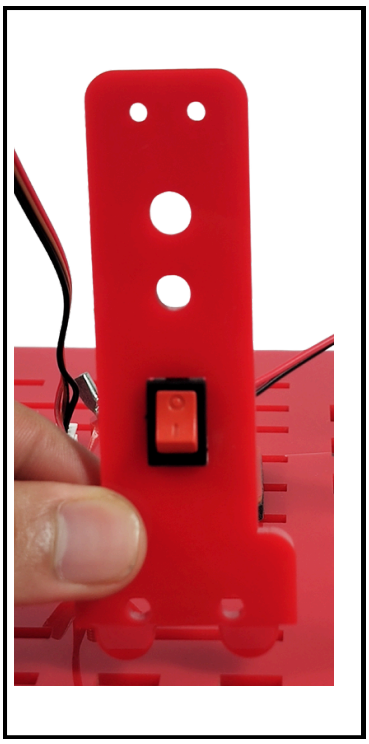


Step 2- Connect the rocker switch and the LED to the PCB

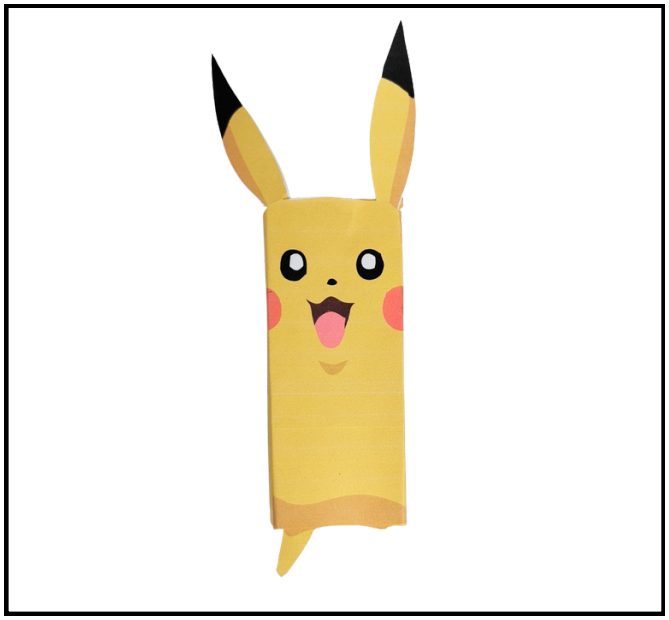


Step 3- Fix the LED and the buzzer to the stands respectively using bolts and attach the whole piece to the board

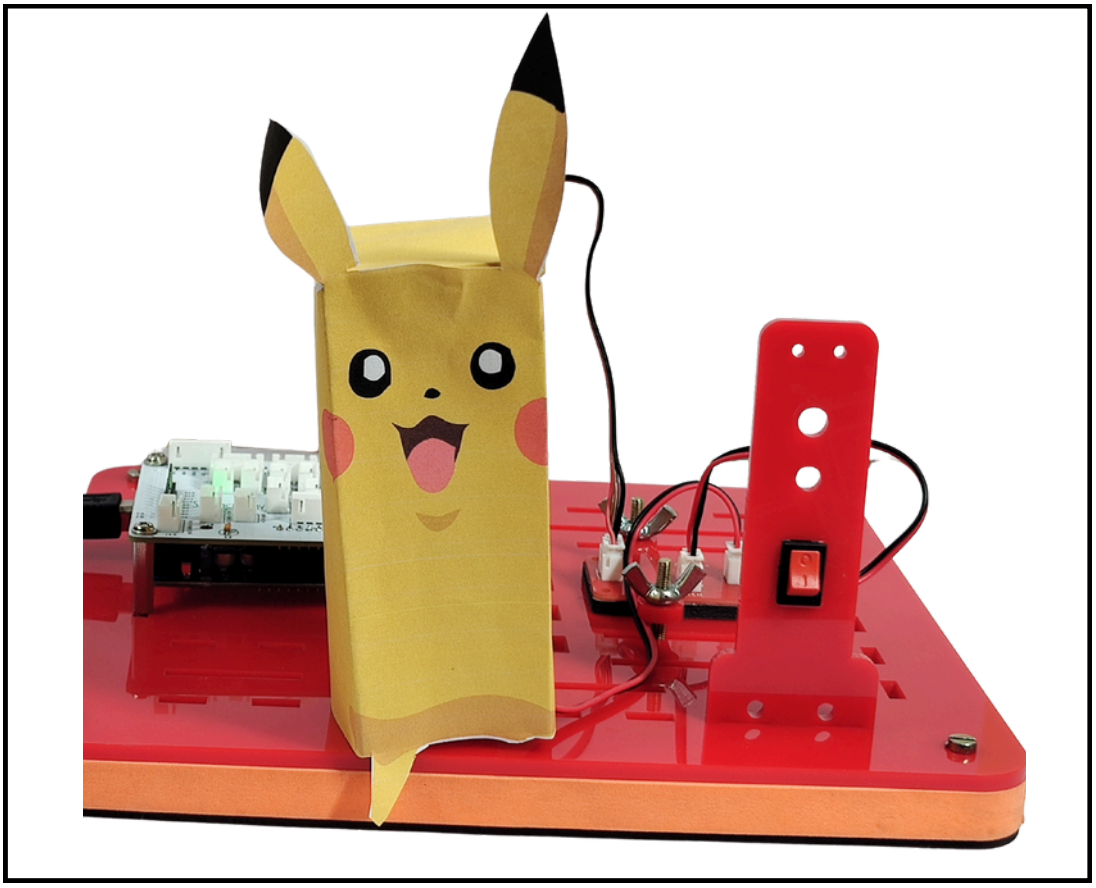




Step 4- Fold the Pikachu as shown below



Step 5- Place the Pikachu on the stand having LED



Your Pikachu is now redy to light up!

code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

const int led = 6;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  digitalWrite(led, HIGH);
  delay(500);
  digitalWrite(led, LOW);
  delay(500);
}
```